

⌈ Preliminary.

⌈ $(\mathbb{Z}_n, +, \cdot)$ is a Ring. If $n=p$ is prime, then for any $a \in \mathbb{Z}_p$, because

$$a^{p-1} \equiv 1 \pmod{p}.$$

That is, $a^{-1} = a^{p-2} \pmod{p} \in \mathbb{Z}_p$.

⌈ Let F be a field. If $\text{ch}(F)=0$, then F must contain $\mathbb{Q}$,

If $\text{ch}(F) \neq 0$, then F must contain $\mathbb{F}_p$, where $p = \text{ch}(F)$

⌈ If $\deg_{\mathbb{F}_p} \alpha = n$, then $F(\alpha)$ has p^n elements.

Because $F(\alpha) \cong \mathbb{F}_p^n$

⌈ If E is finite field, then E has p^n elements for some $n \in \mathbb{N}$, where $p = \text{ch}(E)$.

Because since E is finite, $\dim_{\mathbb{F}_p} E$ must be finite.

⌈ Thm. Let $\mathbb{Z}_p \leq E \leq \overline{\mathbb{Z}_p}$ has p^n elements. Then, for $p(x) = x^{p^n} - x \in \mathbb{Z}[x]$,

$$E = \{\alpha \in \overline{\mathbb{Z}_p} \mid p(\alpha) = 0\}$$

Proof. Since $(E \setminus \{0\}, \cdot)$ is a group of order $p^n - 1$, for $\alpha \in E \setminus \{0\}$, $|\alpha|$ divides $p^n - 1$.

So that $\alpha^{p^n - 1} = 1$, therefore $p(\alpha) = 0$. And, the polynomial $p(x)$ can have at most p^n zeros, consequently we obtain the result.

Thm. Let F be a finite field of characteristic p .

Then, a map

$$\varphi_p: F \rightarrow F: x \mapsto x^p$$

is Automorphism. Moreover, $F_{\{\varphi_p\}} = \mathbb{Z}_p$.

Proof. Well-defined is clear.

Homomorphism: for any $a, b \in F$,

$$\varphi_p(a+b) = (a+b)^p = \underbrace{\sum_{k=0}^p \binom{p}{k} a^{p-k} b^k}_{\text{binomial}} = \underbrace{a^p + b^p}_{\text{ch}(F)=p} = \varphi_p(a) + \varphi_p(b)$$

And clearly preserves multiply.

Injectivity:

$$\varphi_p(a) = a^p = 0 \Leftrightarrow a=0 \Rightarrow \ker \varphi_p = \{0\}.$$

Surjectivity: Finite injective endomorphism, thus surjective.

Meanwhile, by the Fermat's Theorem, for any $a \in \mathbb{Z}_p$, (isomorphically the prime field $\mathbb{Z}_p$)

$$\varphi_p(a) = a^p = a$$

Thus $\mathbb{Z}_p \subseteq F$. And, observe that

$$\varphi_p(x) = x \Leftrightarrow x^p = x \Leftrightarrow x^p - x = 0.$$

This polynomial has at most p zeros in F , thus $\mathbb{Z}_p = F_{\{\varphi_p\}}$.

Corollary. If $\text{ch}(F)=p$, then $(\alpha+\beta)^{p^n} = \alpha^{p^n} + \beta^{p^n}$ for any $\alpha, \beta \in F$, $n \geq 1$.

The existence of $GF(p^n)$.

Thm. If F is a field with $\text{ch}(F)=p$, then $x^{p^n}-x \in F[x]$ has p^n distinct zeros in $\overline{F}$.

Proof. observe that $x^{p^n}-x = x \cdot [x^{p^n-1}-1]$. Let $f(x) = x^{p^n-1}-1$.

Suppose that $\alpha \in \overline{F}$ is a root of $f(x)$. Then

$$f(x) = (x-\alpha) \cdot \underbrace{(x^{p^n-2} + \alpha x^{p^n-3} + \dots + \alpha^{p^n-3}x + \alpha^{p^n-2})}_{p^n-1 \text{ terms}}$$

Let $g(x) = f(x)/(x-\alpha)$. Now,

$$g(\alpha) = [(p^n-1) \cdot 1] \cdot \alpha^{-1} = -\alpha^{-1} \quad (\text{since } (p \cdot 1)x = 0, \text{ ch}(F)=p)$$

But $\alpha \neq 0$, thus $-\alpha^{-1} \neq 0$. This means α is a zero of f of multiplicity 1.

A finite field of order of p^n exists.

Proof. Let $K = \{x \in \overline{\mathbb{Z}_p} \mid x^{p^n}-x=0\} \subseteq \overline{\mathbb{Z}_p}$